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Finite Element Analysis of Human Femur Under Static Load

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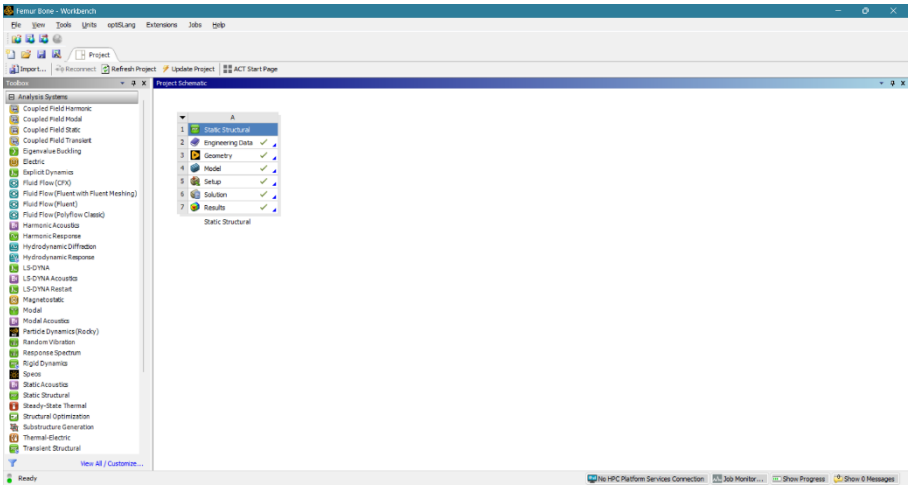
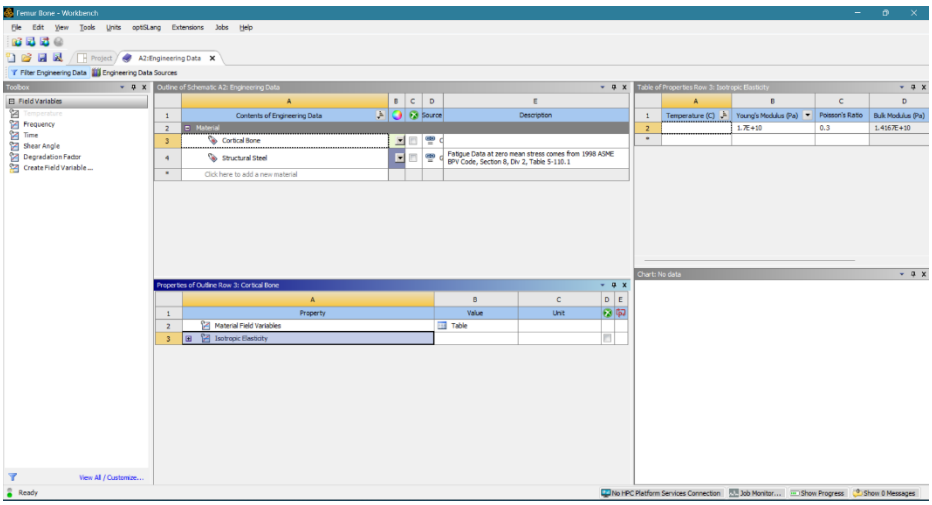
1. Presentation of the analyzed case

In this project, ANSYS Workbench was used to analyze a three dimensional finite element of a human femur. The study aim was to examine the mechanical behavior of the femur under a stationary physiological load which emulated the load imposed on the body through the hip joint. The geometry of the femur was imported based on a CAD model and was an approximate anatomical form of a human femur. There was a statical analysis of the structure with the aim of calculating the deformation, stress and strain distribution in the bone.



2. Material selection with justification

The femur was represented as a straight and homogenous, isotropic, and linear elastic body that represented cortical bone. This is the assumption that is usually taken in biomechanical finite element analysis to simplify the structural analysis.



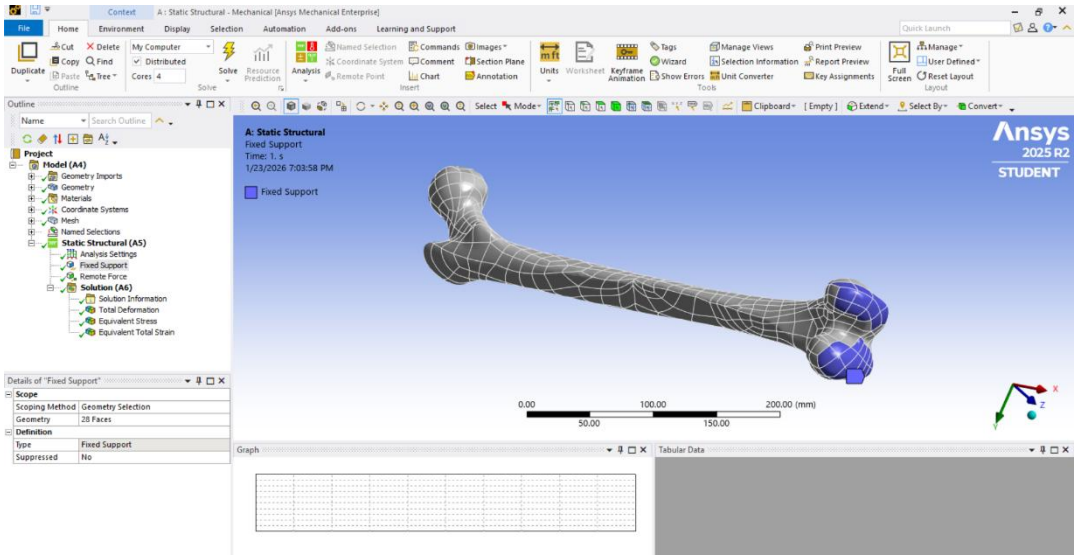
The material properties assigned to the femur were:
Young's modulus: E = 17 GPa
Poisson's ratio: v = 0.3

These values are associated with the normal mechanical values that are found in literature on cortical bone. By morphologically modeling the femur as cortical bone, it is possible to realistically approximate the stiffness and the load carrying behavior of the femur under physiological loading conditions.

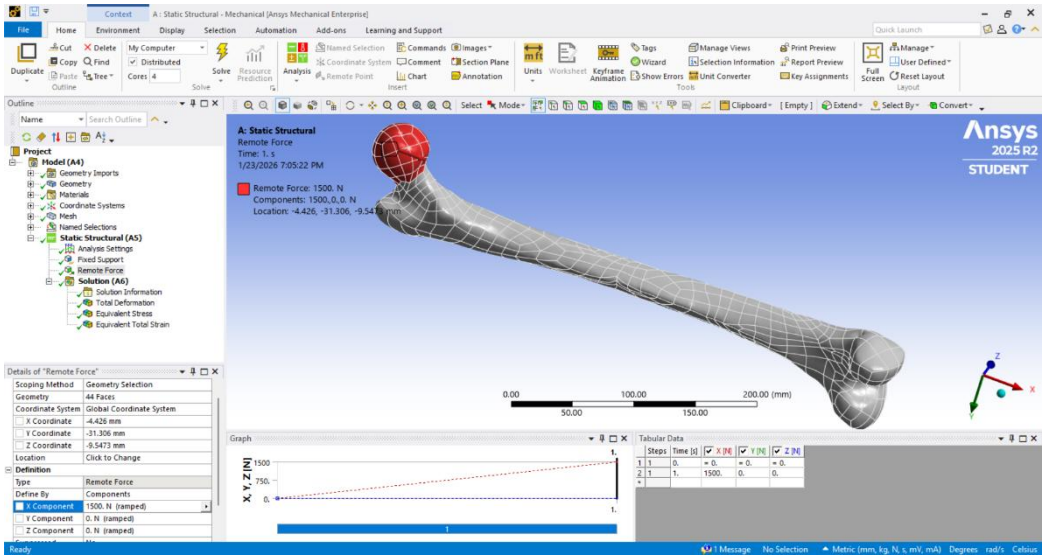
3. Defining boundary conditions and loading

In order to model realistic boundary conditions, the end of the femur (knee region) was fixed by a support which did not allow any translation or rotation. This is the mechanical limitation of the femur at the knee joint. A remote force of 1500 N was applied to the femur at the head. The load was applied on the whole surface of the femur head in order to show the transfer of the load through the hip joint in a standing position. The direction of applied force was parallel to the principal axis of the femur, which created a compressive loading state and a bending loading state as in physiological conditions.

Fixed Support:



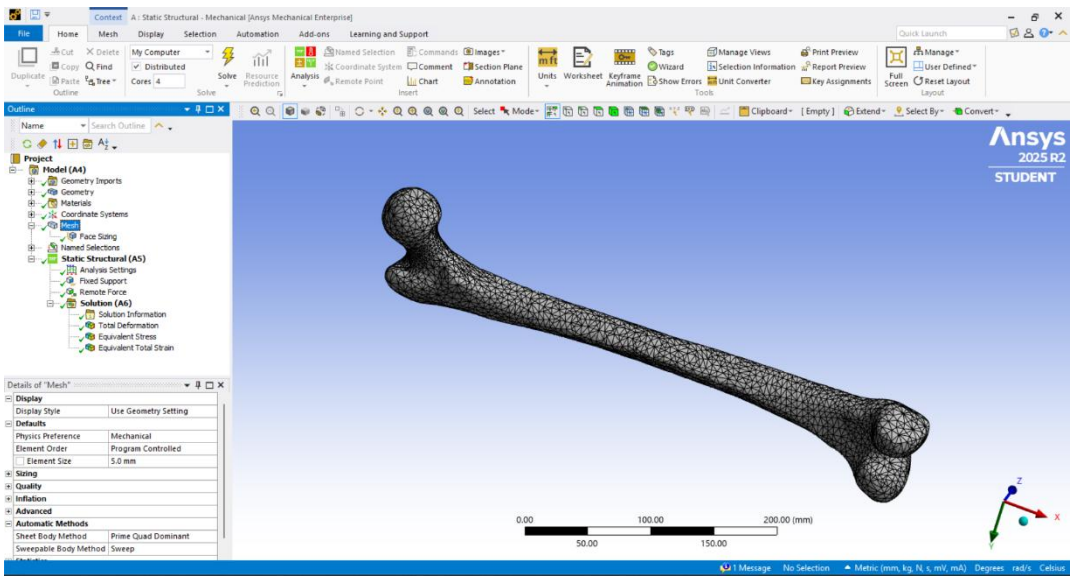
Remote Force:



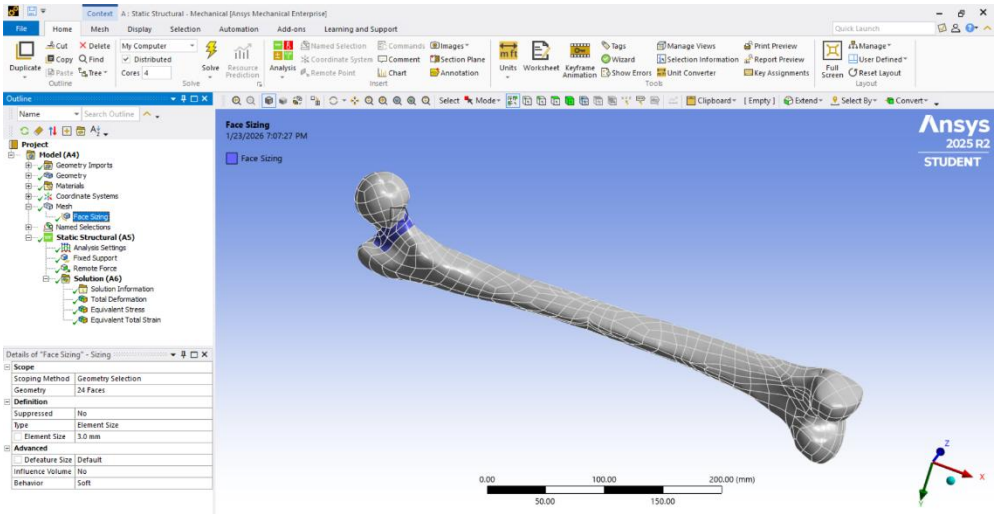
4. Mesh description

A tetrahedral finite element mesh was used to discretize the geometry of the femur. The size of the entire model was set to global element size 5 mm in order to make it computationally efficient. To enhance stress resolution in high stress gradient areas, local mesh refinement was also applied in the femoral head and the neck region with element size of 3 mm. This mesh plan is a good balance between computational cost and numerical accuracy.

5mm mesh:



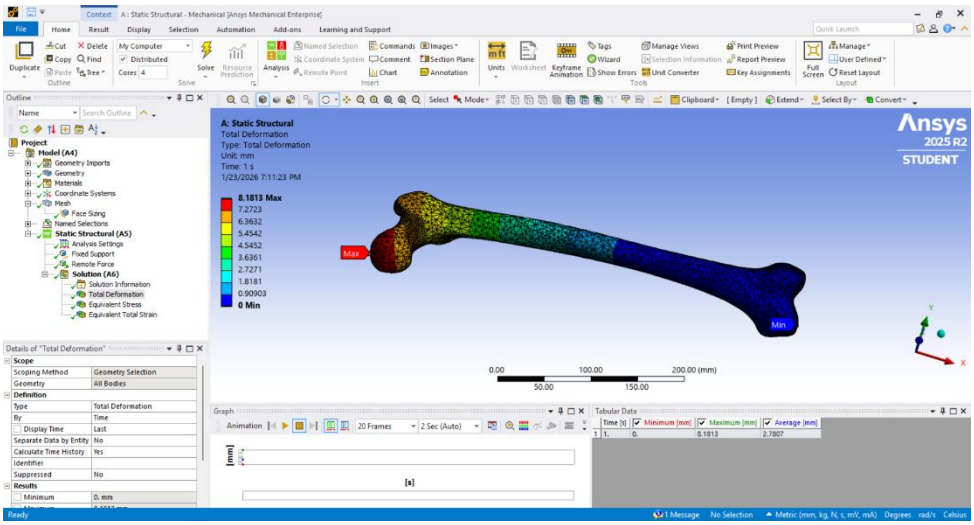
3mm sectional sizing:



5. Results:

a) Total deformation

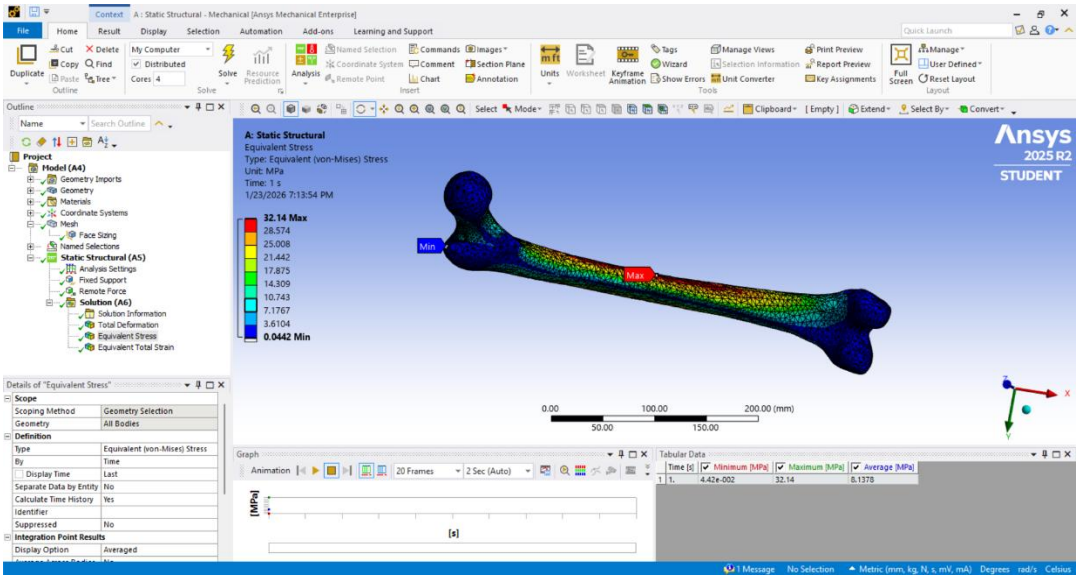
The overall deformation results show that the largest deformation of about 8.2 mm occurs at the femur head where the external force acts. The least deformation is found at the fixed distal end. The pattern of deformation indicates that the femur was bent under the loading and the boundary conditions applied.



b) Equivalent stress (von Mises)

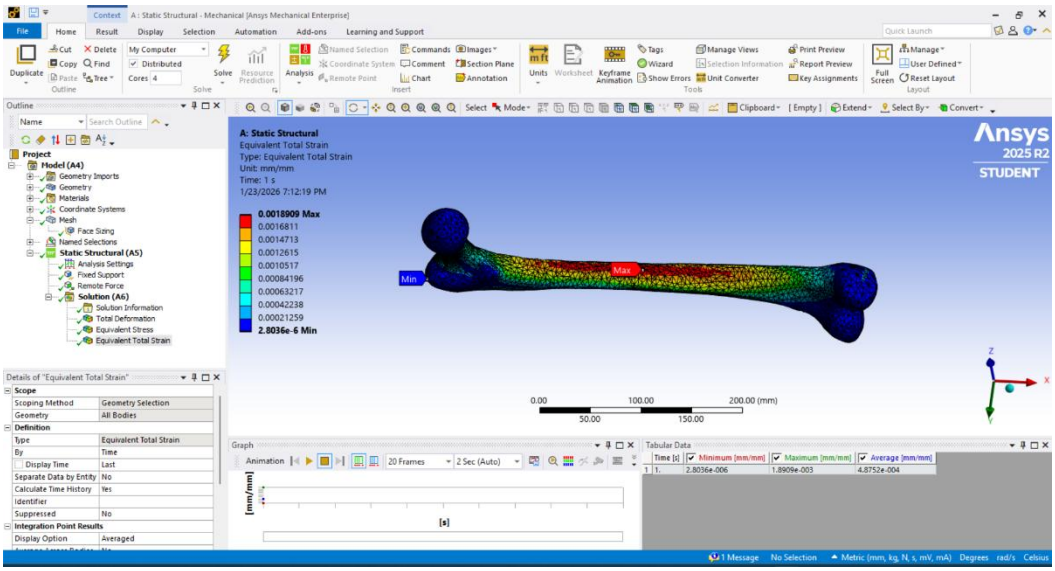
The equivalent von Mises stress distribution indicates that the highest stress of about 32.6 Mpa is concentrated on the neck and the femur shaft. This is the region where the bending moment is the maximum. The resulting values of stress are very low compared to the

ultimate strength of cortical bone (100-150 Mpa) which shows that the femur is working within reasonable elastic parameters under the applied mass.



c) Equivalent strain

The upper limit of the equivalent strain was around 0.0019 (0.19%), which falls within the normal physiological range of strain of cortical bone. This proves that the material is still in the elastic regime and permanent deformation should not be anticipated.



6. Conclusion

ANSYS was used to successful to complete a finite element analysis of a human femur under static loading. The model was able to model realistic deformation, stress, and strain distributions in line with biomechanical behavior in literature. The load applied caused safe

stress on the cortical bone in the elastic range of the bone. Local mesh refinement enhanced accuracy of results in the critical areas with computation cost that was not too high. This analysis confirms the relevance of finite element techniques in biomechanical analysis of bones.